Quick-Start Guide

MS Word Parser is a digital-forensics tool that extracts the hidden evidence inside Microsoft Word files — authorship metadata, editing-session identifiers (RSIDs), comments, tracked changes, protection settings and the structure of the ZIP package itself — and writes it to Excel and/or SQLite for review.

Supported files: `.docx` `.docm` `.dotx` `.dotm` Office lock files (names starting ~$) are skipped automatically.

01 Install

Requires Python 3.9 or later. The graphical user interface (GUI) needs one extra component (PyQt6), so install the `[gui]` option if you plan to use it.

With the GUI (recommended)

```
python3 -m pip install "ms-word-parser[gui]"
```

Command line only

```
python3 -m pip install ms-word-parser
```

Launch the GUI with `parse-docx-gui`. The tool has two separate commands: `parse-docx-gui` for the GUI and `parse-docx` for the command line (`parse-docx -g` also opens the GUI). If `parse-docx-gui` is missing or reports missing dependencies, the `[gui]` option has not been installed. Simply run the above GUI install command, or you can download the standalone GUI executable from the GitHub page. Run `parse-docx -h` to list every command-line option.

02 Choose a processing mode

MODE	BEST FOR	WHAT YOU GET
Triage FAST	A first pass over a large collection (thousands of files) to find documents worth a closer look.	Document Summary, Metadata, Comments and an Excel Tips sheet.
Full DEEP	Files flagged during triage, or any set you need fully parsed. Use SQLite output for large volumes.	Everything in Triage plus RSIDs, Archive Files, Track Changes, comment threads and reactions, People, Document Protection, Range Permissions, Rights Management, Content Types, Headers & Footers, Footnotes & Endnotes, Custom Properties, Item XML and Ink XML.

03 Add options

OPTION	WHAT IT DOES
Hash Files	Records the MD5 hash of each document and of every part inside it. Use it to attest to file integrity later and to spot duplicates.
Timeline	Merges every timestamped artifact (created, modified, last printed, comments, reactions, archive parts, ink) into one chronological view. In Excel you also get a Visual Timeline chart.
To Excel	Writes an .xlsx workbook. Sheets over 1 million rows are split automatically (RSIDs_1, RSIDs_2 …).
To SQLite	Writes a .db database: one table per artifact, indexed, with views for Aggregated Comments and the Timeline. Faster and far smaller for big Full runs.

You can pick Excel, SQLite or both. Every run also writes a log file. All outputs share one name: `ms-word-parser-YYYYMMDD_HHMMSS` with `.xlsx`, `.db` or `.log`.

04 Run it from the GUI

1. Under **Processing Options**, choose **Triage** or **Full**, and tick any of **Hash Files**, **Timeline**, **To Excel** and **To SQLite**.
2. Click **Output Path** and pick the folder where the results and log file will be saved.
3. Add documents in one of three ways: **Add Files** (one or more files in a folder), **Add Directory** (you will be asked whether to load every subfolder too), or **Add Ingest** (a text file of full paths, one per line).
4. Click **Process**. The **Processing Status** panel shows **# of Files**, **# of Errors** and **# Remaining**, with a live message feed.
5. When it finishes, use **Open Log File** or **Open Output Path** to go straight to the results.

Stop

Ends processing early. Results for the files already parsed are still written, along with the log.

Reset

Clears the session so you can choose a new output path and new files for another run.

The **Style** menu changes the interface's appearance, and **Help › Contents** opens built-in guidance.

05 Run it from the command line

A command-line run needs four choices: an output folder, a source of files, a mode and at least one output format.

FLAG	PURPOSE
`-o, --output DIR`	**Required.** Existing folder for results and the log.
`--files F1 F2 … | --dir DIR | --ingest LIST.txt`	**Pick one.** Specific files, a folder, or a text list of full paths.
`-r, --recurse`	With `--dir`, include every subfolder.
`-t, --triage | -f, --full`	**Pick one.** Processing mode.
`-e, --excel / -s, --sqlite`	**At least one.** Output format(s).
`-H, --hash`	Hash each file and every part inside it (MD5).
`-T, --timeline`	Add the Timeline sheet or SQLite view.
`-v / -vv`	Echo progress to the screen (INFO / DEBUG) as well as the log.

Combine flags. Flags that take no value (`-t`, `-f`, `-r`, `-e`, `-s`, `-H`, `-T`) can be grouped behind a single dash, so `-fresHT` is the same as `-f -r -e -s -H -T`. Only one of `-t` or `-f` can be used per run.

Examples

```
# Triage a whole evidence folder, including subfolders, to Excel
parse-docx -tre --dir D:\Evidence -o C:\Cases\Out

# Full parse of a shortlist (ingest file) to Excel and SQLite, hashed, with a timeline
parse-docx -fesHT --ingest shortlist.txt -o C:\Cases\Out
```

While it runs, press **Space** for a progress update or **Q** to quit. Pressing Ctrl+C also stops the run and writes the results gathered so far.

06 Understand the output

Each artifact becomes a worksheet in Excel or a table in SQLite. Sheets only appear when the files contain that data.

SHEET / TABLE	MODE	CONTENTS
Document Summary	T	One row per file: hash, RSID Root, unique rsidR count, tag counts, document IDs, and flags for comments, ink, protection, track changes and encryption.
Metadata	T	Author, last modified by, created / modified / printed dates, revision count, editing time, template, application.
Comments	T	One row per comment or reply: ID, timestamp, author, initials, text.
RSIDs	F	Each RSID type and value with how often it appears — ties text to the save session it was typed in.
Archive Files	F	Every part inside the .docx package: size, timestamp, ZIP attributes and extra-field bytes.
Comment threads	F	Extended, Extensible and Comments IDs: reply structure, resolved status, reactions and who reacted.
Track Changes	F	Tracked insertions and deletions with author and date.
Protection & rights	F	Document Protection, Range Permissions and Rights Management (IRM issuer and validity dates).
Other parts	F	People, Content Types, Headers & Footers, Footnotes & Endnotes, Custom Properties, Item XML, Ink XML.
Errors	Both	Any file that could not be processed, with the reason.

07 Recommended workflow

1. **Triage everything.** Run Triage across the full collection.
2. **Filter.** Sort and filter the Document Summary and Metadata to find documents of interest — for example, matching RSID Root values.
3. **Make a list.** Copy their full paths into a text file, one per line.
4. **Go deep.** Re-run in Full mode using Ingest and a compiled list of file paths. For large datasets, choose SQLite output.

08 Good to know

- **27 ready-made SQLite queries** are available on the project's GitHub repository at `github.com/jjrboucher/MS-Word-Parser/tree/master/sqlite_queries`. They are not installed with the Python package or included in the standalone executables, so download them from there. Examples include files sharing an rsidR value, files last modified by someone other than the author, and low editing time across more than one page. Run them in a tool like DB Browser for SQLite.
- The **Excel Tips** sheet walks through pivot-table techniques, such as grouping documents by RSID Root.
- Timestamps dated before 2007 are treated as invalid and left blank.

Caution: page, line and character counts come from the document's own metadata and can be out of date. Re-saving a file to correct them changes its modified date and author — always work on copies of evidence.